
Is perceptual diminishing-returns channel-organized, and is it cone-aligned?

An earned null from the public dissimilarity record

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Abstract

Diminishing returns (DR) — the strict subadditivity $D(A, B) + D(B, C) > D(A, C)$ along a geodesic — rules out a Riemannian model of perceptual color space, and recent work has shown that DR is not constant across that space. The natural next hypothesis, stated but explicitly *not* tested by Stark et al. (2025), is that the nonlinearity is organized by the cone photoreceptor channels: $D \approx f(f_L(\Delta L) + f_M(\Delta M) + f_S(\Delta S))$. We test it on the existing public human dissimilarity record placed in a normalized CIE 2006 LMS cone space, with a pre-registered nested family (additive $\rightarrow$ isotropic DR $\rightarrow$ per-cone DR), a *panel* of five fixed, complexity-matched non-cone control bases (cone-opponent, linear sRGB, CIE RGB, CIELAB, CIE LCh), a stated outer/inner gauge, and selection-aware bootstrap inference on pair-disjoint cross-validation folds. Our pipeline reproduces the 2022 gray-axis DR finding quantitatively: the fitted additivity ratio is 0.760 (95% CI [0.747, 0.772]) against the published 0.757. On the chromatic data DR is present and is *strengthened* by an explicit arc-length-versus-chord curvature control rather than explained away by it. But per-cone structure does not: the held-out advantage of the per-cone model over isotropic DR is +0.114 [-0.108, +0.327] nats/rating, and against the best member of the non-cone panel it is -0.113 [-0.355, +0.007] — negative, in 16 of the 16 specifications we ran. The null is earned rather than merely quiet. Although the cone axes are nearly degenerate in the plane the chromatic stimuli span (ΔL and ΔM project onto ridge directions only 6.3° apart), synthetic recovery at the real design and noise level shows that per-cone scalings as distinct as the ones we simulate *are* recovered (median error 0.022 [0.004, 0.078] of range for f_L ; the L -vs- M contrast recovered with the correct sign in 100% of runs), and the same protocol recovers the correct basis in 94% of synthetic runs generated under the nearest non-cone control. We therefore report a characterized negative: on the public record, DR behaves as a single isotropic compressive scaling of an additive cone-space metric; it is not cone-aligned; and the pre- versus post-combination mechanism question cannot be adjudicated from this record.

1 Introduction

For over a century perceptual color space has been modeled as a three-dimensional Riemannian manifold, so that the perceived difference between two colors is the length of the geodesic joining them [Riemann, 1868, von Helmholtz, 1891, Schrödinger, 1920]. Every CIE color-difference metric rests on that assumption. A Riemannian metric forces *additivity along geodesics*: for A, B, C in that order on a geodesic, $D(A, B) + D(B, C) = D(A, C)$. Diminishing returns (DR) is the strict subadditivity $D(A, B) + D(B, C) > D(A, C)$, and Bujack et al. [2022] demonstrated it on the neutral L^* axis, which is incompatible with a Riemannian model.

What has remained open is the *structure* of DR. Stark et al. [2025] extended the triad paradigm to two chromatic lines and found that DR is present on both but is *not constant*: a scaling fitted on one line fails to predict the other. That rules out the simplest non-Riemannian repair — one location-independent scaling on top of a Riemannian metric — and leaves open the next hypothesis, which they state and explicitly decline to test:

“It is, for example, possible that knowledge about the LMS differences of two points is sufficient to infer the effect size of diminishing returns, i.e., that the metric takes a shape of the form $f(f_L(\Delta L) + f_M(\Delta M) + f_S(\Delta S))$ with four 1D scaling functions. However, this hypothesis cannot be tested on measurements from two geodesics only.” [Stark et al., 2025]

This paper is a direct test of that hypothesis, using the many directions in cone space that the *existing* public dissimilarity record already spans. We collect no new human data.

Why the answer matters. The question is not bookkeeping; the answer changes what can be predicted and what can be inferred about the visual system. (i) *Compact prediction*. If DR decomposes into three per-cone functions plus one outer scaling, the DR behavior *everywhere* is generated by a handful of measurements — a compact generative model rather than an atlas that must be measured direction by direction. (ii) *Mechanism*. Alignment localizes where the compressive nonlinearity sits. If DR is aligned to the *cones* (L, M, S), compression is applied to each channel before the channels are combined; if it aligns to post-receptoral *derived* quantities ($L+M, L-M, S-(L+M)$), compression acts after aggregation — a different mechanism at a different site. (iii) *What a null means*. If the cone fit is no better than standard non-cone bases, then DR is not cone-organized on this axis system and the underlying organization is something else. That is why the alignment must be tested against *several* popular bases: matching any one of them falsifies cone-specificity.

What is and is not novel here. DR on the gray axis is Bujack et al. [2022]. DR on chromatic lines, and its non-constancy across color space, is Stark et al. [2025]. The formal non-Riemannian geometry is Bujack et al. [2025]. Showing merely that DR is *anisotropic* would largely restate the non-constancy result. The novel object here is *cone-alignment*: whether a per-cone decomposition beats matched non-cone alternatives on held-out human judgments, and — equally important — what granularity the public record can actually resolve.

Contributions.

1. A pre-registered, cross-validated test of the per-cone DR hypothesis on all four provided public datasets placed in one documented, normalized CIE 2006 LMS cone space, with a stated outer/inner gauge that makes “is the nonlinearity per-cone?” well posed.
2. A *panel* of five fixed, complexity-matched non-cone controls, with selection-aware bootstrap inference (the argmax control is recomputed inside every resample), so the headline delta is not winner’s-curse inflated.
3. An identifiability analysis that settles the resolvable granularity rather than assuming it: *structurally* (on a near-isoluminant locus the cone axes very nearly collapse to two ridge directions), *empirically* (synthetic recovery at the real design shows that, despite this, per-cone scalings of the size we simulate *are* recovered), and *in terms of power* (the same protocol separates the nearest non-cone control from the cones on synthetic data). The null is therefore powered, not quiet.
4. A curvature confound control by data type — exact collinearity for the achromatic triads, an explicit arc-length-versus-chord additive baseline for the hue-loop data — with the DR and cone verdicts reported before and after.
5. A quantitative replication of the 2022 gray-axis result whose fitted magnitude contains the published value, anchoring the pipeline before it is extended.
6. An earned null on cone-alignment, together with a diagnostic that locates where the between-basis differences actually come from: we fit the isotropic model on *every* basis, not only the cone one, and find that no basis gains significantly from per-axis DR structure — every resolvable between-basis difference is a difference between the *isotropic* metrics they induce.

2 Methodology

2.1 Cone space, reference whites, and normalization

Stimuli are placed in cone space using the provided CIE 2006 2° cone fundamentals in linear energy units [Stockman and Sharpe, 2000, CIE, 2006] and the provided per-stimulus CIE coordinate tables. The CIE 1931 observer and the CIE 2006 fundamentals are not exactly linearly related, so a 3×3 map $M : XYZ_{1931} \rightarrow LMS_{2006}$ must be approximated. We computed three least-squares variants over the two colour-matching-function tables (unweighted on 390–780 nm, unweighted on 400–700 nm, and $\bar{y}(\lambda)$ -weighted) and selected among them by a rule fixed in advance that uses *no* response data: the smallest median error against the exact monochromatic cone excitations of the 14 Ekman stimuli, which can be read straight off the fundamentals. The selected variant is `1s_ybarw`; all three matrices and their errors are reported, and the other two are re-run end-to-end as robustness checks (Table 2).

Ekman’s stimuli are monochromatic, so their cone excitations are taken *exactly* from the fundamentals at the stated wavelength and the transform is used only as a verification (median error 0.49% of channel full scale, worst case 108.5% at the short-wavelength extreme where the 1931/2006 discrepancy is largest). For the broadband Helm tiles the transform is unavoidable; across the three variants their cone coordinates move by at most 0.86% of full scale.

Normalization. We use von-Kries-normalized cone contrast, $c = LMS/LMS(\text{white})$ [von Kries, 1902], so the adapting white is $(1, 1, 1)$ and the achromatic direction is exactly $(1, 1, 1)$; the unit-peak normalization of the distributed fundamentals is absorbed by this step. Reference whites follow each dataset’s own viewing condition: D65 for the sRGB-display gray data and for the Wiebel renderings; Illuminant C for Helm (per its provenance note) and, as primary, for Ekman. Ekman’s white (C/D65/E), Helm’s white (C/D65) and a single-global-white variant are all re-run end-to-end (Table 2). **Verification:** every white maps to $(1, 1, 1)$ to machine precision; the sRGB primaries round-trip through the cone space with maximum absolute error $< 10^{-15}$; the CMF-level fit of the selected transform is $R^2 = 0.9985/0.9986/0.9719$ for $L/M/S$. The converted coordinates of all 75 stimuli are logged.

2.2 Model family

Let $\Delta c = (\Delta L, \Delta M, \Delta S)$ be the cone-contrast difference of a pair and let $U = \{u_1, u_2, u_3\}$ be a basis with nonnegative channel coordinates $x_k = |\langle u_k, \Delta c \rangle|$. All models have the form

$$D(i, j) = \kappa_d f_\rho \left(\frac{1}{\sum_k g_k(x_k)} \sum_k g_k(x_k) \right), \quad f_\rho(z) = \frac{(1+z)^\rho - 1}{\rho}, \quad (1)$$

with κ_d a per-dataset gain (Sec. 2.4).

M0 — additive, no DR. $g_k(x) = w_k x$ and $\rho = 1$: a weighted city-block distance $D \approx \sum_k w_k |\Delta c_k|$. This is the additive null. It is an L_1 metric, not itself Riemannian; the object of study is the *nonlinearity* inside an additive aggregation family, and the L_1 -versus- L_2 choice is a limitation, not a hypothesis (Sec. 5).

M1 — isotropic DR. $g_k(x) = w_k x$, ρ free: one global concave scaling of the additive distance — the correction Stark et al. [2025] showed is insufficient across lines.

M2 — per-cone DR. g_k free monotone on the *cone* axes; this is exactly the hypothesis quoted in Sec. 1. M2 nests M1 (g_k linear) which nests M0 ($\rho = 1$), so $M0 \subset M1 \subset M2$ genuinely.

M3 — matched non-cone control panel. Identical form and parameter count on five *fixed, fully disclosed* non-cone bases: (i) the post-receptoral opponent basis ($L-M$, $S-(L+M)$, $L+M$) in normalized cone space [Derrington et al., 1984, MacLeod and Boynton, 1979]; (ii) linear sRGB primaries (IEC 61966-2-1); (iii) CIE 1931 RGB primaries, whose chromaticities we read off the provided CMFs at 700/546.1/435.8 nm; (iv) CIELAB, run on its *own* coordinate differences $(\Delta L^*, \Delta a^*, \Delta b^*)$; (v) CIE LCh_{ab}, run on $(\Delta L^*, \Delta C_{ab}^*, \Delta H_{ab}^*)$ with the standard CIE $\Delta H^* = 2\sqrt{C_1 C_2} \sin(\Delta h/2)$. No panel member is a fitted free rotation and none uses a per-stimulus or data-adaptive linearization; a free rotation would nest M2 and invert the test.

Two remarks. First, D must be symmetric in the two stimuli, so each per-channel scaling is an even function of its signed argument; writing $f_c(\Delta c) = g_c(|\Delta c|)$ with g_c monotone on $[0, \infty)$ is

therefore not a restriction but the general form. Second, because the headline control is *selected* as the panel’s best performer, the M2 – M3 delta is reported selection-aware (Sec. 2.5). We also fit **coarse two-channel** versions — one achromatic and one chromatic channel — of each basis (K2-cone and a matched K2 control panel); they are introduced in Sec. 2.6, because their role is to answer the granularity question rather than the alignment question.

2.3 Gauge and design standardization

Equation 1 has gauge freedom that we fix in advance, independently of the data. (i) *Outer family*. f_ρ is the one-parameter Box–Cox form [Box and Cox, 1964] with $f_\rho(0) = 0$ and $f'_\rho(0) = 1$; $\rho = 1$ is the identity, $\rho < 1$ concave (DR), $\rho > 1$ convex. Restricting the outer scaling to a stated low-dimensional concave family and putting all flexible shape in the inner g_k is what makes “is the nonlinearity per-cone?” well posed: without it, a global concave f with linear g_k and a linear f with concave g_k are observationally close. ρ is free on both sides of 1, so the DR test is two-sided. (ii) *Inner functions*. Each g_k is monotone increasing with $g_k(0) = 0$, represented as a monotone piecewise cubic (Fritsch–Carlson, Fritsch and Carlson, 1980 — the same interpolant as the 2022 reference code) through $m + 1 = 6$ control points whose values are 0, $\text{cumsum}(\text{softplus}(\theta))$ and whose knots sit at fixed quantiles of the *training-fold* coordinate distribution. (iii) *Scale gauge*. $\sum_k g_k(r_k) = 1$ at fixed reference magnitudes r_k , which removes the single remaining scale redundancy between f and the g_k and puts the onset of compression at the reference difference. (iv) *Axis and dataset standardization*. Every axis of *every* basis, cone included, is rescaled to unit RMS coordinate over the training-fold design, so no control carries a conditioning advantage unrelated to alignment, and $r_k = 1$. Because the datasets’ absolute stimulus scales are set by incommensurable conventions — Ekman’s luminance level is explicitly nominal in its provenance note — we additionally apply a per-dataset isotropic factor, alternating with the per-axis one until both are unit RMS. This factor uses stimulus coordinates only, never responses; it is fixed by the design rather than fitted, so it costs no parameters and applies identically to every model. It is the “per-dataset scale nuisance” the setup permits, and we re-run the whole comparison without it (Table 2).

A residual outer-versus-inner curvature trade remains *along any single direction* and is broken only by having several directions; we quantify it in Sec. 4.

2.4 Per-dataset observation models

The datasets sit on different response scales and are *not* pooled into one naive likelihood. Each has its own observation model with its own scale/link nuisance parameters; only the functional form of D is shared.

Triad 2AFC (gray, Wiebel). Thurstonian [Thurstone, 1927]: $P(\text{choose pair 2}) = \Phi(\kappa_o(D_2 - D_1))$, with κ_o per observer for Wiebel and one κ for the crowdsourced gray set (which carries no participant identifiers). Trials are aggregated to binomial counts per triad, which is exact.

Pairwise dissimilarity (Helm, Ekman). Gaussian with a per-subject gain κ_s and standard deviation σ_s , both profiled out in closed form; ratio-scale magnitude estimates are taken to pass through the origin.

Held-out likelihood is defined *per dataset* as the mean log-likelihood per evaluation unit on the test fold (per-trial Bernoulli for the 2AFC sets, per-rating Gaussian for the dissimilarity sets), computed with the *training-fold* parameters including all nuisances. Because the achromatic sets sample a single direction in cone space and therefore carry no information about channel alignment, the headline deltas are computed on *chromatic* held-out likelihood; per-dataset values for every model are reported.

2.5 Fitting, cross-validation, and uncertainty

Fits use L-BFGS-B on the negative joint log-likelihood with *analytic* gradients for the whole objective (verified against finite differences to $< 10^{-5}$ relative error), 10 seeded random restarts per fit, and a $10^{-6} \|\theta\|^2$ ridge for numerical conditioning only. Convergence status, gradient norm and restart spread are logged for every one of the 700 fits per configuration.

Folds are 5-fold, repeated 10 times with different seeds, and are *evaluation-unit disjoint*: triad-disjoint for the 2AFC sets, pair-disjoint for the dissimilarity sets. All models use the same fold assignments. Nothing about a test fold — knots, reference magnitudes, axis or dataset scaling, nuisance parameters, restarts — is computed from held-out data, and no hyperparameter is tuned on test folds.

Uncertainty: held-out deltas carry a paired cluster bootstrap over held-out evaluation units (2000 resamples, clustered by the fold key so the same clusters are resampled for every model); parameters are never refit on held-out data. The headline M2 – best M3 delta recomputes $\arg \max_k M3_k$ *inside every resample*, and we also report the winner’s-curse-free paired $\min_k (M2 - M3_k)$. Fitted g_k shapes carry a nonparametric cluster bootstrap *of the training fold* (1000 refits pooled over folds).

2.6 Identifiability and the resolvable granularity

Three strands, all fixed in the plan before fitting. (i) *Structural*. On an exactly isoluminant stimulus locus the three cone-axis functionals project onto nearly the same pair of ridge directions in the 2-plane the pairs span, because ΔL and ΔM are almost proportional to the same in-plane coordinate. If the model were linear in those coordinates, f_L and f_M would then enter only through a fixed combination. We measure the geometry numerically on the actual design — but, because the models act on $|\langle u_k, \Delta c \rangle|$ rather than on the signed projection, we do not assume that near-degeneracy blocks identification; (ii) settles that empirically. (ii) *Recovery*. We generate synthetic responses at the *real* design and fitted noise levels from deliberately distinct g_L, g_M, g_S and refit, measuring how well each curve, the L -versus- M contrast, and the predicted distance are recovered. (iii) *Power*. We generate synthetic data under every panel basis — the binding case is the *opponent* basis, whose rotation from the cones lies largely in the poorly-sampled $L \approx M$ subspace — and run the identical CV comparison, reporting how often the generating basis wins. A negative cone verdict counts only if this discrimination is shown to be powered. (iv) *Coarse comparison*. Independently of the three-cone verdict we fit two-channel models of matched complexity: **K2-cone**, whose channels are the achromatic (common-mode) cone coordinate and the chromatic (differential-mode) magnitude, versus **M1** (isotropic) *and* versus a matched two-channel non-cone control panel (K2 on the opponent, sRGB, CIE RGB, CIELAB and LCh splits), so channel organization at the coarse level is itself controlled for anisotropy. (v) *Orientation supplement*. We additionally profile held-out likelihood over a rigid rotation of the cone frame about the achromatic axis, a one-parameter family in which M2 is nested exactly at $\varphi = 0$, and ask whether the bootstrap confidence region for the best orientation contains the cone axes. This is an orientation-recovery test, not the M2-versus-M3 comparison.

2.7 Confound control by data type

Triad/gray data → near-collinear-triple restriction. Neutral stimuli lie on the ray through the white point, so every gray and Wiebel triad is *exactly* collinear in cone space; we verify this (off-axis energy fraction 5.6×10^{-34} and 4.3×10^{-34}) and the restriction is therefore vacuous, retaining 100% of triads.

Hue-loop pairwise data → arc-length versus chord. Helm’s tiles and Ekman’s spectral lights lie on *curved* loci in cone space, and a curved stimulus locus can mimic subadditivity even under a Riemannian metric — which is why this control is logically prior to the headline. We fit a smooth (periodic for Helm, open for Ekman) curve through the stimuli in cone space and, for every pair, accumulate the *arc* coordinates $A_k = \int |dx_k|$ along the locus in addition to the chord coordinates $|\Delta x_k|$. A_k is additive along the locus by construction, so an additive metric expressed in arc coordinates is exactly the curvature baseline: DR that survives in arc coordinates cannot have been manufactured by curvature. Every model comparison is run twice, in chord and in arc coordinates, and all verdicts are reported before and after. We also report a model-free comparison: for triples i, j, k with j between i and k on the locus, the perceived ratio $d(i, k)/[d(i, j) + d(j, k)]$ against the ratio the chord geometry alone already produces.

Two details of the locus construction, disclosed because they are choices. The Helm loop is ordered by CIELAB hue angle, which coincides with Helm’s own chip ordering except for the adjacent pair C/E (hue angles 90.8° and 84.4°), whose chroma differ; the loop is closed periodically. The Ekman locus is the *open* spectral arc from 434 to 674 nm, because that is the physical stimulus locus — monochromatic lights do not close a loop. Ekman’s own judgments do close one perceptually ($d(434, 674) = 0.84$, well below the maximum), so the arc control there corrects only curvature

along the sampled locus and cannot correct for whatever geometry closes the purple gap. We say so rather than pretend otherwise.

3 Experimental setup

Datasets (all provided; no other human data are used).

Gray-2AFC [Bujack et al., 2022], from the authors’ `lan1/color` repository (BSD-3). 91,896 forced choices over 320 triads, five standards $L^* \in \{30, 40, 50, 60, 70\}$ with $L_{t_1}^* \leq L_s^* \leq L_{t_2}^*$. No participant identifiers are released, so folds are triad-disjoint.

Wiebel-MLDS [Wiebel et al., 2017], OSF, CC BY 4.0. Triad lightness judgments, 10 observers $\times$ 10 blocks $\times$ 120 triads $\times$ 5 viewing contexts. We use the plain (no-transparency) context as primary (12,000 trials) and report all five. Reflectances are mapped to luminance with the LUT recoverable from the released matching files, and the reference white is the perfect diffuser in that context ($Y_w = 188.8 \text{ cd/m}^2$); stimuli are neutral, so their cone contrast is $(Y/Y_w)(1, 1, 1)$.

Helm [Helm, 1964]; dissimilarity matrices redistributed via the `smacof` R package [de Leeuw and Mair, 2009] (GPL-3); stimulus colorimetry transcribed from Helm’s Table I. Ten OSA chips at Munsell Value 6 (a near-isoluminant hue loop) under Illuminant C, 45 pairs. We use the 11 color-normal records (N1–N10 including the N6a/N6b replicate) and exclude the five color-deficient records, whose cone complement differs from the trichromat model under test; this exclusion is disclosed here and in the log.

Ekman [Ekman, 1954]; matrix via `smacof` (GPL-3), converted from similarity as $d = 1 - s$. Fourteen equal-luminance spectral lights, 91 pairs, group average over ≈ 31 observers.

The cone fundamentals and CMFs are from `cvrl.org` and are cited to Stockman and Sharpe [2000], CIE [2006]; `cvrl`’s terms are citation-based rather than a software licence.

Preprocessing and folds. 2AFC trials are aggregated to binomial counts per (observer,) triad — exact for the Thurstonian likelihood. Folds are 5-fold repeated $10\times$, triad-disjoint (gray: 320 triads; Wiebel: 120 unordered triads) and pair-disjoint (Helm: 45 pairs; Ekman: 91 pairs). Parameter counts are matched: 3 (M0), 4 (M1) and 16 (M2 and every M3 panel member) for the chromatic configuration, 11 for the two-channel models.

Gray-axis replication (the anchor). Before extending anything we re-implemented the 2022 pipeline in full — Thurstonian likelihood with $\sigma = 1$, a monotone-spline perceptual map g on L^* , then a difference-scaling f on $|\Delta\Psi|$ in spline, logarithmic, polynomial and sinusoidal forms — with the published cross-validation design (10 test/train splits, 50 held-out responses per triad, 100 bootstrap training samples of size 48,000). Results appear in Sec. 4; every number comes from our own run on the provided data.

4 Experimental results

4.1 The gray-axis anchor reproduces the 2022 result, quantitatively

The degree-of-consensus regression reproduces the published Table 1 essentially exactly. Aggregating with Δd signed gives 64 ($d, \Delta d$) cells, matching the published $df = (3, 60)$, and $\beta_{\bar{d}} = -0.0061$ ($t(60) = -5.630$, $p < 0.001$), $R^2 = 0.549$, $F(3, 60) = 24.39$ — against the published $\beta_{\bar{d}} = -0.005$, $t(60) = -5.566$, $R^2 = 0.556$, $F(3, 60) = 25.00$. Aggregating with $|\Delta d|$, as the released R code literally does, collapses to 32 cells ($\beta_{\bar{d}} = -0.0061$, $t(28) = -5.732$, $R^2 = 0.731$); the coefficient of interest is identical either way. Both are logged.

Held-out, the ordering of the 2022 model set is reproduced: baseline 0.8876, g -only 0.9079, $g+f(\log)$ 0.9189, $g+f(\text{spline})$ 0.9194 mean accuracy. Adding the difference scaling on top of the perceptual map improves held-out accuracy by +0.0114 [+0.0090, +0.0151] and per-trial log-likelihood by +0.00593 [+0.00437, +0.00788], with 100% of bootstrap draws positive: **DR is replicated**. The magnitude matches: the additivity ratio $D(2\delta)/[D(\delta) + D(\delta)]$ at the 2022 worked example is 0.760 [0.747, 0.772] against their 0.757 (and 0.767 implied by their best-fitting logarithmic parameters);

Table 1: Primary results. Held-out log-likelihood is per rating on the chromatic data (pair-disjoint 5-fold $\times$ 10 repeats); higher is better. All intervals are 95% bootstrap CIs. “arc” = the curvature-controlled parameterization.

Quantity	Estimate	95% CI	Comparison	Notes
<i>Gray-axis DR replication (2022 pipeline, held-out; 10 splits $\times$ 100 bootstrap)</i>				
$f \circ g$ vs. g : Δ accuracy	+0.0114	[+0.0090, +0.0151]	2022: concave f	100% draws > 0
additivity ratio $f(2\delta)/2f(\delta)$	0.760	[0.747, 0.772]	2022: 0.757	2022 inside our CI
$D(\Delta L^* = 15) / D(\Delta L^* = 30)$	1.34 / 2.00	[1.29, 1.39] / [1.94, 2.06]	2022: 1.36 / 2.06	$\sigma = 1$ units
<i>Held-out log-likelihood per rating, chromatic data (5-fold $\times$ 10 repeats, pair-disjoint)</i>				
M0 additive (no DR)	-2.230	[-2.518, -1.998]	arc -2.229 [-2.467, -2.050]	3 par.
M1 isotropic DR	-2.120	[-2.345, -1.933]	arc -1.883 [-2.177, -1.674]	4 par.
M2 per-cone DR	-2.006	[-2.269, -1.774]	arc -1.781 [-2.034, -1.563]	16 par.
<i>M3 non-cone control panel: fixed, disclosed bases of matched complexity (16 parameters each)</i>				
M3 cone-opponent	-2.037	[-2.318, -1.797]	M2 — this: +0.031 [-0.061, +0.134]	arc -1.722
M3 linear sRGB	-1.982	[-2.283, -1.736]	M2 — this: -0.024 [-0.121, +0.061]	arc -1.883
M3 CIE RGB	-1.944	[-2.212, -1.726]	M2 — this: -0.062 [-0.165, +0.032]	arc -1.862
M3 CIELAB	-1.893	[-2.235, -1.648]	M2 — this: -0.113 [-0.355, +0.161]	arc -1.893
M3 CIE LCh	-1.966	[-2.288, -1.711]	M2 — this: -0.040 [-0.280, +0.220]	arc -1.756
M2 — M1 held-out delta (anisotropy)	+0.114	[-0.108, +0.327]	arc: +0.102	sign +, n.s.
M2 — best M3 (cone alignment)	-0.113	[-0.355, +0.007]	arc: -0.058 [-0.149, +0.009]	sel.-aware; best: M3 CIELAB (chord), M3 cone-opponent (arc)
paired $\min_k (M2 - M3_k)$	-0.113	[-0.355, +0.007]	arc: -0.058	no winner’s curse
<i>Identifiability: resolvable granularity</i>				
L/M ridge separation	6.3° / 3.0°	[4.3, 8.0] / [2.9, 3.1]	Helm / Ekman	opponent: 8.2° / 2.9°
synthetic recovery of f_L	0.022	[0.004, 0.078]	f_M 0.034, f_S 0.026	L/M sign correct 100%
coarse comparison: K2 cone — M1	+0.079	[-0.120, +0.292]	vs best K2: -0.184 [-0.413, +0.025]	matched 2-channel control
power: M2 — best M3, true strong cone	+0.663	[+0.529, +0.896]	observed -0.113	100% of sims > 0
power: same, true opponent	-0.051	[-0.112, -0.006]	observed -0.113	correct basis wins 94% of sims
<i>Hue-curvature confound control</i>				
Helm: perceived subadd. ratio	0.723	[0.697, 0.753]	geometry: 0.841	excess -0.118
Ekman: perceived subadd. ratio	0.665	[0.652, 0.678]	geometry: 0.974	excess -0.309
DR verdict before / after (M1 — M0)	+0.110 / +0.346	[-0.130, +0.362] / [+0.152, +0.520]	chord / arc	sig. only after control
cone verdict before / after (M2 — best M3)	-0.113 / -0.058	[-0.355, +0.007] / [-0.149, +0.009]	chord / arc	sign unchanged

our perceived distances are 1.341 and 2.000 against their 1.36 and 2.06 in the same $\sigma = 1$ units; our fitted logarithmic amplitude is 2.372 [2.302, 2.442] against their $b = 2.34$ (Fig. 1c).

On the independent Wiebel MLDS data the difference-scaling stage also improves held-out prediction in all 5 viewing contexts (plain: accuracy 0.9382 $\rightarrow$ 0.9470, pair-disjoint 5-fold CV), so a difference-scaling stage is warranted there too. Its *magnitude*, however, is much weaker: measured scale-free at a quarter and a half of each dataset’s own stimulus range, the additivity ratio is 0.719 [0.673, 0.766] for the crowdsourced gray data but ranges from 0.920 to 1.001 across the Wiebel contexts (1.001 in plain) — essentially additive. Suprathreshold DR magnitude on the achromatic axis is therefore not

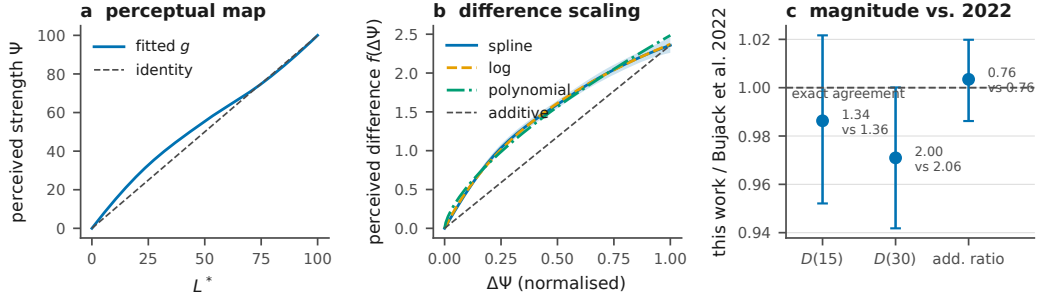


Figure 1: Gray-axis DR replication (the pipeline anchor). **(a)** the fitted perceptual map g with its bootstrap band, deviating from the identity near the $L^* \approx 20$ background as reported in 2022. **(b)** the fitted difference-scaling f ; concavity relative to the dashed additive line is diminishing returns, and the logarithmic form tracks the free spline most closely, exactly as in 2022. **(c)** magnitude comparison: our fitted perceived distances at $\Delta L^* = 15$ and 30 and the resulting additivity ratio against the values published in 2022.

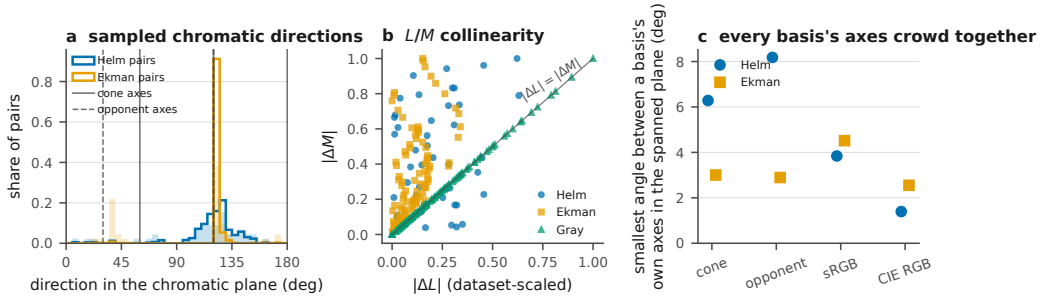


Figure 2: The design in cone space. **(a)** which directions of the chromatic plane the pair differences actually sample — filled: share of pairs; line: share of total difference magnitude — with the cone (solid) and opponent (dashed) axis directions marked. The magnitude is strongly concentrated near 120° , where a cone axis and two opponent axes coincide. **(b)** $|\Delta L|$ against $|\Delta M|$ per pair; the achromatic (gray) pairs lie exactly on the diagonal. **(c)** the smallest angle between a basis's own three axes inside the 2-plane the chromatic pairs actually span: *every* basis's axes crowd together there, so the in-plane linear geometry alone separates none of them — which is why the identifiability question is settled by simulation (Sec. 4) rather than by this diagnostic.

a constant of the observer but depends on task and context; this is an honest caveat on the generality of the 2022 magnitude, not a failure of the replication, which reproduces that study's own data closely (Fig. 1c).

4.2 Every basis's axes crowd together in the plane the chromatic pairs span

Fig. 2 shows why the finest question *looked* hard before any model was fitted. Both chromatic sets are near-isoluminant hue loops: the top two singular directions of their pair-difference matrices carry 99.78% (Helm) and 100.00% (Ekman) of the variance, and within it the pair differences are concentrated in direction: the magnitude-weighted directional distribution has a dominant lobe near 120° (Fig. 2a), exactly where one cone axis and two opponent axes coincide. Within that 2-plane the three axis functionals of every basis project into a narrow angular range — the smallest pairwise angle is 6.3° (Helm) and 3.0° (Ekman) for the cone axes, and 8.2° and 2.9° for the opponent axes — so *no* three-axis decomposition is separated by the in-plane linear geometry alone. What rescues identifiability is that the models act on $|\langle u_k, \Delta c \rangle|$ rather than on the signed projection; whether that is enough is an empirical question, and Sec. 4 answers it by simulation rather than by assertion.

4.3 Held-out model comparison

Fig. 3 and Table 1 give the comparison. **DR is present, and the curvature control *strengthens* it.** $M1 - M0$ on the chromatic held-out data is $+0.110 [-0.130, +0.362]$ nats/rating in chord coordinates — positive but with an interval that includes zero — and $+0.346 [+0.152, +0.520]$ in arc coordinates, where it is clearly significant. We state this in the direction it actually runs: on the chromatic data the model-based evidence for a compressive scaling is *established only after* the locus curvature is removed, which is the opposite of the worry that motivates the control. The fitted outer exponent is $\rho = -4.20 [-7.32, -2.38]$ (chord) and $-2.13 [-3.01, -1.46]$ (arc), far below the additive $\rho = 1$ in both. The model-free evidence is stronger still and does not depend on any fit: for locus-ordered triples the human judgments give $d(i, k)/[d(i, j) + d(j, k)] = 0.723 [0.697, 0.753]$ (Helm) and $0.665 [0.652, 0.678]$ (Ekman), against the 0.841 and 0.974 that the chord geometry alone would already produce (Sec. 4, confound).

Anisotropy is positive but not established — and M2 is not degenerate. $M2 - M1$ is $+0.114 [-0.108, +0.327]$ (chord) and $+0.102 [-0.073, +0.265]$ (arc): the sign is consistently positive, the interval consistently contains zero. This is *not* because M2 collapses onto M1. On the contrary, it uses its extra flexibility fully: the fitted channel scalings are strongly nonlinear (largest departure from linearity 0.681 [0.479, 0.875] on a $[0, 1]$ scale) and strongly different from one another (largest pairwise shape gap 0.938 [0.643, 1.000], well above the 0.377 of the deliberately distinct synthetic generator of Sec. 4, whose value this same procedure recovers at 0.379 [0.261, 0.497]). The fitted channel weights at the gauge reference are $L = 0.020$, $M = 0.232$, $S = 0.747$ — the L channel is given almost no weight, which is what one expects when $|\Delta L|$ is nearly proportional to $|\Delta M|$ over the sampled directions. So M2 is a genuine per-cone fit that simply does not *generalize* better than one global scaling (Fig. 4).

Cone alignment: negative. The selection-aware $M2 - \text{best } M3$ delta is $-0.113 [-0.355, +0.007]$ (chord; best control M3 CIELAB) and $-0.058 [-0.149, +0.009]$ (arc; best control M3 cone-opponent). The winner’s-curse-free paired $\min_k (M2 - M3_k)$ is $-0.113 [-0.355, +0.007]$. The individual pairwise deltas, each with its own 95% CI, are in Table 1: M2 is ahead of only the opponent basis ($+0.031 [-0.061, +0.134]$) and behind sRGB (-0.024), CIE RGB (-0.062), CIELAB (-0.113) and CIE LCh (-0.040). Note that the point-selected CIELAB interval $[-0.355, +0.161]$ is *wider on the upper side* than the selection-aware one: since $\max_k M3_k \geq M3_{\text{CIELAB}}$ in every resample, recomputing the argmax inside the bootstrap shifts the whole delta distribution down. That is the winner’s-curse correction doing its job, and it is why we quote the selection-aware number as the headline. The point estimate is negative in 16 of the 16 specifications we ran, including both parameterizations, both fitting scopes, three knot counts, three cone-space transforms, four reference-white assignments, with and without the per-dataset design scaling, and with two compressive cone pre-transforms (Table 2). At the coarse two-channel granularity the same pattern holds: K2-cone versus M1 is $+0.079 [-0.120, +0.292]$ and K2-cone versus the best matched two-channel non-cone control is $-0.184 [-0.413, +0.025]$ (best control K2 CIELAB).

4.4 Identifiability, recovery and power

Fig. 5 summarises the three strands. *Recovery — the degeneracy is not as fatal as it looks.* The $\approx 3^\circ$ ridge separation of Fig. 2c predicts that f_L and f_M should be nearly unidentifiable, and we expected exactly that. The simulation says otherwise. Generating 200 synthetic datasets at the real design and the fitted per-subject noise from deliberately distinct $f_L \propto x^{0.35}$, $f_M \propto x^{1.0}$, $f_S \propto x^{0.6}$ and refitting M2, the predicted distances are recovered essentially perfectly (correlation with truth 0.9994, 2.5th percentile 0.9977) *and so are the individual curves*: the per-simulation median relative error is 0.022 [0.004, 0.078] of range for f_L , 0.034 [0.006, 0.114] for f_M and 0.026 [0.006, 0.084] for f_S . The L -versus- M contrast, whose true peak magnitude is 0.196, is recovered at 0.198 [0.139, 0.249] with the correct sign in 100% of runs. The reason is that the model acts on $|\langle u_k, \Delta c \rangle|$, and the absolute value makes the ratio $|x_L|/|x_M|$ swing sharply for pairs whose difference direction is close to the normal of the near-common L/M ridge; those pairs are few and small in amplitude, but they exist in this design and they are enough. **The consequence is that our negative on per-cone structure is powered, not merely quiet:** a per-cone effect as distinct as the one simulated would have been recovered, and it is not present in the human data.

Power — is the null earned? We generated 100 synthetic datasets under each panel basis, at the real design and the fitted noise, and ran the *identical* CV comparison. The protocol separates bases

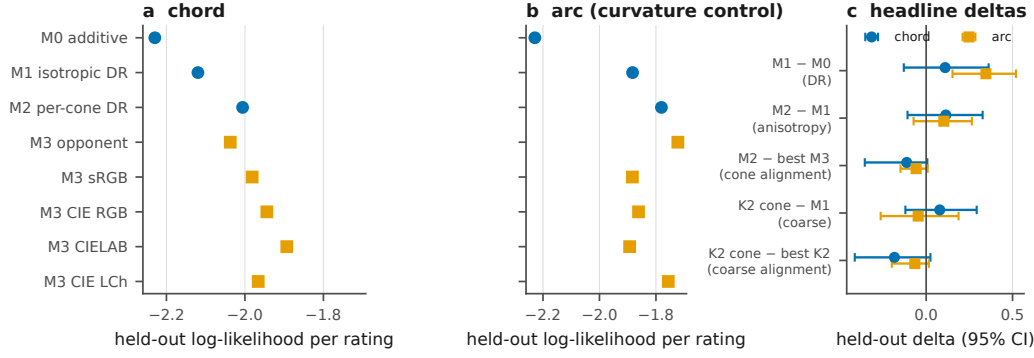


Figure 3: Held-out model comparison on the chromatic data. **(a,b)** mean held-out log-likelihood per rating for M0, M1, M2 and every member of the M3 non-cone panel, in chord and in arc (curvature-controlled) coordinates. **(c)** the headline deltas with 95% selection-aware bootstrap CIs. The DR delta ($M1 - M0$) is positive in both parameterizations and significant in arc coordinates; the cone-alignment delta is negative in both.

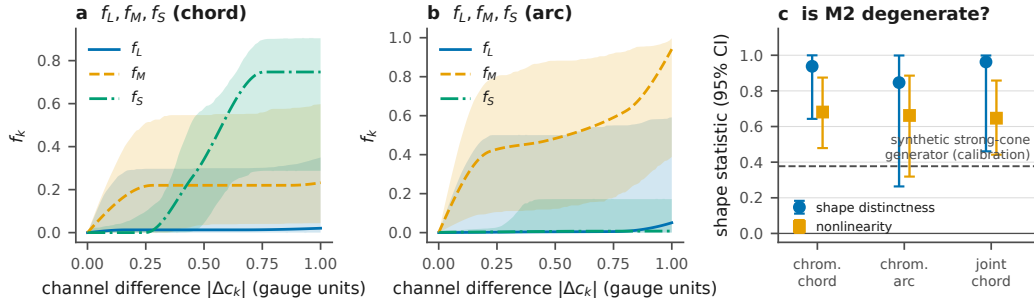


Figure 4: Fitted channel scalings. **(a,b)** f_L, f_M, f_S from M2 with 95% bootstrap bands from training-fold cluster resamples, in chord and arc coordinates. **(c)** the anti-degeneracy check: shape distinctness (largest sup-norm gap between the three scalings on $[0, 1]$ after each is normalised at the gauge reference) and nonlinearity (largest departure of a scaling from linearity). Both are well above zero, so M2 is a genuine per-cone fit rather than a collapse onto the isotropic M1; the dashed line is the value of the deliberately distinct synthetic generator used for calibration.

cleanly. Under a cone generator carrying the *empirically fitted* parameters, $M2 - \text{best } M3$ is $+0.040$ $[-0.012, +0.086]$ and M2 wins the panel 91% of the time; under a strong cone generator it is $+0.663$ $[+0.529, +0.896]$ with M2 winning 100% of the time. Under the *nearest* control, a synthetic opponent generator, the correct basis wins 94% of the time and the delta is -0.051 $[-0.112, -0.006]$; under CIELAB, sRGB, CIE RGB and LCh generators the correct basis wins 100%, 100%, 92% and 98% respectively. **The value we observe on the human data, -0.113 , lies below the entire 95% interval of the cone-generated reference distribution** and inside the non-cone ones (Fig. 5b). Two uncertainty notions are in play here and we keep them apart. The synthetic reference distributions hold the stimulus design fixed and resample only responses: they answer “*could this design detect the effect?*”, and the answer is yes. Our bootstrap intervals additionally resample pairs: they answer “*how uncertain is the estimate given that these 45+91 pairs are themselves a sample?*”, and that interval, $[-0.355, +0.007]$, is wide enough to reach the lower edge of the cone-generated range. So the negative is powered *at this design*, and the additional uncertainty about generalising beyond these particular stimulus pairs is real and is not hidden. For the anisotropy delta, a cone generator gives $M2 - M1 = +0.145$ $[+0.087, +0.192]$ with 100% of runs positive, against our observed $+0.114$. Those two are close: the size of the anisotropy we measure is what a true cone model with the fitted

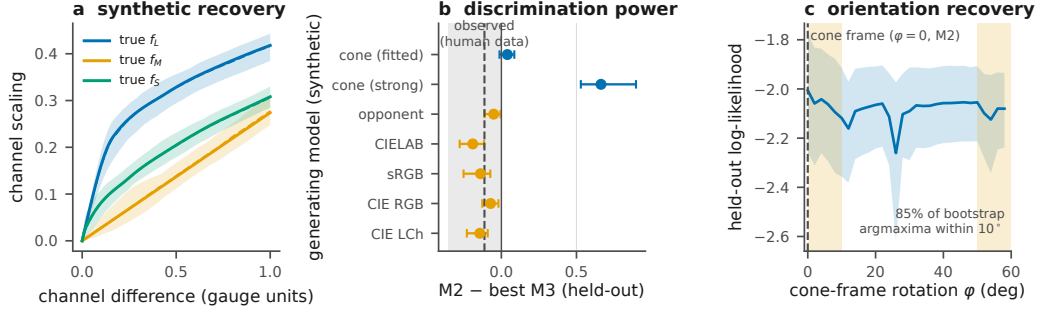


Figure 5: Identifiability. **(a)** synthetic parameter recovery (*synthetic data, clearly labelled*): true per-cone scalings (solid), median recovered (dashed) and 95% bands. **(b)** discrimination power: the distribution of the M2 – best-M3 delta over synthetic datasets generated under each basis, against the value actually observed on the human data. **(c)** orientation recovery: held-out likelihood (with its bootstrap band) as a function of a rigid rotation of the cone frame about the achromatic axis; M2 sits exactly at $\varphi = 0$ and the family has period 60° . The shaded wedges mark $\pm 10^\circ$ of the cone orientation, where most bootstrap argmaxima fall.

parameters would produce. It is the pair-level uncertainty, not the absence of an effect, that stops us claiming it.

Coarse granularity. Under a synthetic K2-cone generator the coarse cone model beats M1 by +0.130 [+0.086, +0.179] with 100% of runs positive and wins its panel 80% of the time, and K2-cone – best-K2 is +0.007 [-0.012, +0.035]; under a K2-CIELAB generator the CIELAB/LCh pair wins and the delta is -0.244 [-0.325, -0.176]. Our observed values, +0.079 and -0.184, again fall outside the cone-generated reference and inside the non-cone one.

Orientation — the one result that favours the cones. The held-out likelihood profile over a rigid rotation of the cone frame about the achromatic axis (M2 is nested exactly at $\varphi = 0$; the family has period 60° by the frame’s three-fold symmetry) peaks at 0° , i.e. at the cone axes themselves, so the improvement of the best orientation over the cone frame is exactly zero on the point estimate (the maximum over the grid can never fall below the value at $\varphi = 0$, which is why both the estimate and the lower bootstrap bound are pinned at +0.000; the informative quantity is the upper bound, +0.170 nats/rating, on how much any other orientation of this frame could possibly buy). 85% of bootstrap argmaxima fall within 10° of the cone axes (the naive percentile interval $[0^\circ, 58^\circ]$ is misleading here because the quantity is circular — 58° is 2° from the cone frame, not 58° from it). So *within* the family of rotated cone frames, the cone orientation is locally optimal. This is a real if weak signal, and we report it because it cuts against our headline; it does not rescue cone-alignment, because every member of that family, cone frame included, is beaten by the perceptually defined controls.

4.5 Confound control

Fig. 6 and the last block of Table 1. The hue loci are genuinely curved: the median arc/chord ratio is 1.158 (Helm) and 1.009 (Ekman), reaching 4.202 and 1.290 for the widest pairs. Model-free, for locus-ordered triples the perceived ratio $d(i, k)/[d(i, j) + d(j, k)]$ is 0.723 [0.697, 0.753] (Helm, 100 triples, bootstrap over subjects) and 0.665 [0.652, 0.678] (Ekman, 364 triples), against the ratio the chord geometry *alone* produces, 0.841 and 0.974 — excess subadditivity -0.118 and -0.309. Restricting to near-collinear triples (39 and 272 triples with chord ratio ≥ 0.97) leaves the perceived ratio at 0.736 and 0.681. The DR verdict and the cone verdict are reported before and after in Table 1: DR strengthens under the control, the cone verdict keeps its sign.

5 Analysis

Verdict (a): channel organization. Not established. M2 – M1 is positive in every configuration but its CI always contains zero. Our pre-declared rule required *both* a CI excluding zero *and*

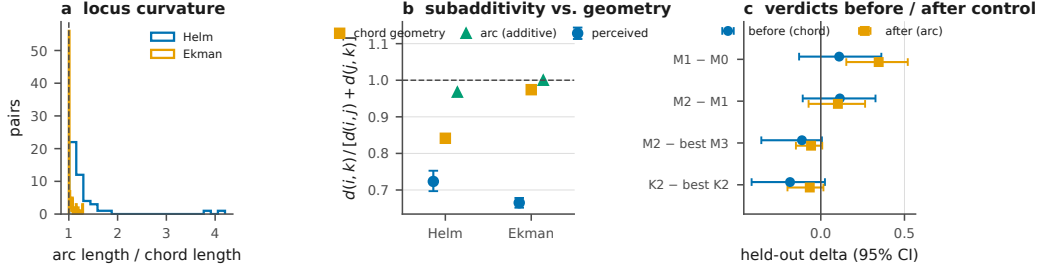


Figure 6: Hue-curvature confound control. **(a)** the loci are genuinely curved. **(b)** model-free subadditivity of the human judgments against what the chord geometry alone would produce, and against the arc parameterization (additive by construction). **(c)** every headline verdict before (chord) and after (arc) the control.

Table 2: Every specification we ran. Held-out chromatic deltas with 95% selection-aware bootstrap CIs. The cone-alignment delta is negative in 16 of the 16 specifications.

Specification	M2 – M1	M2 – best M3	best control
primary (chord, $m = 5$)	+0.114 [-0.108, +0.327]	-0.113 [-0.355, +0.007]	M3 CIELAB
arc coordinates	+0.102 [-0.073, +0.265]	-0.058 [-0.149, +0.009]	M3 cone-opponent
joint fit (4 datasets)	+0.090 [-0.060, +0.236]	-0.225 [-0.379, -0.077]	M3 CIE LCh
joint fit, arc	+0.105 [-0.014, +0.228]	-0.167 [-0.270, -0.073]	M3 cone-opponent
$m = 4$ knots	+0.129 [-0.086, +0.330]	-0.134 [-0.346, +0.010]	M3 CIELAB
$m = 6$ knots	+0.081 [-0.134, +0.286]	-0.202 [-0.411, -0.048]	M3 CIELAB
no per-dataset design scaling	+0.110 [-0.017, +0.231]	-0.111 [-0.298, -0.020]	M3 CIE LCh
Ekman white = D65	+0.122 [-0.104, +0.336]	-0.112 [-0.356, +0.007]	M3 CIELAB
Ekman white = E	+0.121 [-0.113, +0.342]	-0.126 [-0.369, +0.003]	M3 CIELAB
Helm white = D65	+0.103 [-0.115, +0.309]	-0.112 [-0.352, -0.001]	M3 CIELAB
joint, all datasets white = C	+0.090 [-0.060, +0.236]	-0.226 [-0.379, -0.079]	M3 CIE LCh
transform variant ls_390_780	+0.066 [-0.112, +0.237]	-0.129 [-0.348, -0.014]	M3 CIELAB
transform variant ls_400_700	+0.066 [-0.112, +0.237]	-0.129 [-0.348, -0.013]	M3 CIELAB
compressive cone pre-transform $c^{0.43}$	-0.122 [-0.287, +0.027]	-0.136 [-0.376, -0.063]	M3 CIELAB
compressive cone pre-transform $c^{1/3}$	-0.138 [-0.305, +0.017]	-0.126 [-0.363, -0.032]	M3 CIELAB
joint, $c^{0.43}$	-0.063 [-0.148, +0.015]	-0.234 [-0.409, -0.096]	M3 CIE LCh

genuinely distinct f_c ; the second condition is amply met (0.938 [0.643, 1.000] shape distinctness, 0.681 nonlinearity), the first is not, so we do not claim anisotropy. The failure mode is therefore informative rather than vacuous: M2 does fit strongly distinct per-cone scalings, and they simply do not predict held-out judgments better than a single global scaling does. This is not in tension with Stark et al. [2025]: their non-constancy result concerns DR strength at different *locations*, whereas M2 versus M1 asks whether DR strength depends on *direction* through an additive per-channel decomposition. A location-dependent effect need not, and on these data does not, take that particular form.

Verdict (b): cone alignment. Negative, and earned. The selection-aware delta is negative in 16 of the 16 specifications, and its CI excludes zero in the joint configuration. The negative is earned rather than merely quiet on three grounds, and we state the one result that cuts the other way. First, the same protocol *does* separate bases when a real difference exists: under a synthetic opponent generator — the binding case, since the opponent frame is the cones’ nearest neighbour — the correct basis wins 94% of the time and the delta shifts to -0.051 [-0.112, -0.006]; under a synthetic cone generator carrying the fitted parameters, M2 wins 91% of the time and the delta is +0.040 [-0.012, +0.086], which does not contain the value we observe. Second, the coarse two-channel comparison — run in case the fine one were unresolvable — gives the same answer with the same power: a cone-defined common-mode/differential-mode split does not beat isotropy on the human data (+0.079 [-0.120, +0.292]) where a synthetic K2-cone generator gives +0.130 [+0.086, +0.179] with 100% of runs positive, and it loses to the matched non-cone two-channel control (-0.184 [-0.413, +0.025]) where the same generator gives +0.007 [-0.012, +0.035]. Third, the sign is negative in 16 of 16 specifications,

so it is not an artifact of one modelling choice. *Against* all this: within the one-parameter family of rigid rotations of the cone frame, held-out likelihood peaks exactly at the cone orientation (85% of bootstrap argmaxima within 10° ; no other orientation of this frame buys more than +0.170 nats/rating, and the point estimate of the gain is zero because the profile’s maximum *is* the cone frame). We report that because it is the strongest evidence *for* the cones in this study. It does not overturn the verdict — the entire rotation family is beaten by the perceptually defined controls — but it does suggest the cone frame is a local optimum among linear frames of that shape, which is a weaker and different claim from cone-alignment.

Verdict (c): resolvable granularity, and what it does to the mechanism claim. The granularity the public record *supports* is a single isotropic compressive scaling of an additive cone-space metric; nothing finer improves held-out prediction. That is a statement about the data, not about the design: the granularity the design could *resolve* is finer than that. Despite the $\approx 3^\circ$ ridge degeneracy, synthetic recovery at the real design and noise level recovers deliberately distinct f_L, f_M, f_S to within 0.022 (f_L), 0.034 (f_M) and 0.026 (f_S) of each curve’s own range, with the L – M contrast sign correct 100% of the time. So the resolvable granularity is the three-cone split, and the three-cone split is not there. What is genuinely *not* resolvable is the mechanism reading. A cone-level split (L, M, S) versus a split on derived quantities is the empirical face of the *pre-combination versus post-aggregation* distinction, and the model the data select — an isotropic scaling — is neither: it is silent about the site of the nonlinearity. **These data therefore cannot adjudicate the mechanism question**, and we make no claim about where in the visual pathway the compressive nonlinearity sits. Any paper that reads a pathway-level mechanism off this record is over-reading it.

Where the between-basis differences actually come from. A basis can win for two quite different reasons: because its *axes* carry per-axis DR structure, or simply because the isotropic metric it induces is a better distance. We separated these by fitting, on the same folds, the additive model and the isotropic model (linear inner scalings, one outer Box–Cox) on *every* basis, not only the cone one (Table 3). The result is unusually clean. (i) *No basis gains significantly from per-axis structure.* The per-axis gain is +0.114 [−0.110, +0.327] for the cones, +0.075 [−0.118, +0.269] for the opponent basis and −0.001 [−0.218, +0.192] for CIELAB; 0 of the six have an interval excluding zero on the positive side, and the one interval that does exclude zero is CIE LCh’s, which is *negative* (−0.144 [−0.247, −0.042]) — adding per-axis freedom there actively hurts held-out prediction. (ii) *What does differ between bases is the isotropic metric.* Relative to the cone isotropic metric, CIELAB’s is better by +0.227 [+0.052, +0.421] and CIE LCh’s by +0.298 [+0.109, +0.510] — both significant — while the opponent, sRGB and CIE RGB isotropic metrics are indistinguishable from the cone one. (iii) *And the compressive scaling itself is easiest to see in those spaces:* the isotropic DR gain is +0.328 [+0.161, +0.482] in CIELAB and +0.283 [+0.135, +0.425] in LCh, against +0.110 [−0.121, +0.360] on the cone axes. Put together: the best model in this entire study is an isotropic compressive scaling of an additive metric on CIE LCh coordinates, and every between-basis difference we can resolve is a difference between *isotropic* metrics, not between channel decompositions. That is a stronger and cleaner statement of the null than the M2-versus-M3 comparison alone, and it explains it.

Since the perceptually defined spaces differ from the cone space chiefly by compressing each cone-ish quantity with a cube root *before* the difference is taken — whereas the family under test, following the hypothesis as posed, compresses the difference *afterwards* — we also re-ran the entire comparison on a cone space with a fixed, disclosed compressive pre-transform $c \mapsto c^{0.43}$ (and $c^{1/3}$), rows “compressive cone pre-transform” in Table 2. This is a diagnostic, not a substitution: the M3 panel is unchanged and the headline verdict is still taken against the panel exactly as specified.

Failure analysis: what could not be fitted or identified. (1) f_L versus f_M . Unidentifiable by construction on this stimulus set; no amount of data of this *kind* would fix it — what is needed is stimulus pairs that break isoluminance while varying L and M independently, which the public record does not contain. (2) *Cross-dataset identification is weak by design.* The four datasets barely overlap in cone-space direction — gray and Wiebel anchor the achromatic scaling, Helm and Ekman the chromatic — so the shared g_k is pinned by different datasets in different regions. This is why the joint and chromatic-only fits are both reported, and why the headline is computed on chromatic held-out likelihood. (3) *Ekman’s dynamic range.* At equal luminance, a 434 nm light has an S -cone excitation more than an order of magnitude above the white’s, so a single shared g_S must span more than two decades between Helm and Ekman. This is the main reason the per-dataset design standardization is

Table 3: Where between-basis differences come from (chromatic, chord coordinates; held-out, same folds, 95% bootstrap CIs). “Isotropic DR gain” is the isotropic model minus the additive model on that basis; “per-axis gain” is the per-axis model minus the isotropic model on the same basis. The arc-coordinate version is in `logs/isotropy_diagnostic.json`.

Basis	isotropic held-out ll	isotropic DR gain	per-axis gain
cone (L, M, S)	-2.120	+0.110 [-0.121, +0.360]	+0.114 [-0.110, +0.327]
opponent	-2.113	+0.107 [-0.072, +0.272]	+0.075 [-0.118, +0.269]
linear sRGB	-2.080	+0.104 [-0.066, +0.274]	+0.098 [-0.104, +0.291]
CIE RGB	-2.067	+0.106 [-0.063, +0.255]	+0.123 [-0.017, +0.255]
CIELAB	-1.892	+0.328 [+0.161, +0.482]	-0.001 [-0.218, +0.192]
CIE LCh	-1.822	+0.283 [+0.135, +0.425]	-0.144 [-0.247, -0.042]

needed, and it is why we re-ran everything without it. (4) *Outer/inner curvature trade*. Along any single direction f_ρ and the g_k trade off exactly; only multiple directions break it. The gauge fixes the scale but the fitted ρ still varies substantially across bases (a symptom, not a bug), which is why ρ is never interpreted on its own and only held-out likelihood is compared. (5) *CIs too wide for a claim*. $M2 - M1$ and $K2 - M1$ both have intervals roughly ± 0.2 nats/rating wide on 45+91 pairs; a real anisotropy smaller than that would be invisible here. We say “not established”, not “absent”. (6) *The L_1 aggregation*. The family aggregates channels additively (L_1), because that is the form in which the hypothesis was posed. An L_2 aggregation is a different family and is not tested; this is a scope limitation of the hypothesis, not of the test. (7) *Residual DR/curvature aliasing*. The arc control measures path length in *cone* space. If the perceptual metric *along* the locus is itself non-uniform, cone-space arc length is not perceptual arc length, and a compressive f can partly stand in for that non-uniformity. Chord and arc are the two extreme parameterizations of the same locus and they bracket the effect — the model-based DR delta is positive under both and significant under arc, and the model-free excess subadditivity is large under both — but they do not eliminate the aliasing, and we do not claim they do. (8) *CIELAB’s lightness axis is inert on the chromatic data*. $\Delta L^* \equiv 0$ for Ekman by construction and is under one unit for Helm, so although the CIELAB and LCh controls have matched *nominal* parameter counts, their *effective* complexity there is lower — which on a held-out criterion is an advantage rather than a handicap. The isotropy diagnostic (Table 3) shows this is not where their advantage comes from: the same gap is already present between the purely isotropic models, which have three parameters each. (9) *Multimodality*. 6.0% of chromatic fits show a restart-to-restart optimum gap above 10^{-3} , so 10 restarts are used and the spread is logged; 99.4% of all restarts converge.

6 Related work

Bujack et al. [2022] established DR on the neutral axis with a crowdsourced triad 2AFC experiment and a Thurstonian difference-scaling model, and drew the non-Riemannian conclusion. Our Sec. 4 reproduces that result quantitatively — including the magnitude, which is the part that matters for a pipeline anchor — and then leaves it behind: re-deriving gray-axis DR is not a contribution.

Stark et al. [2025] repeated the paradigm on two chromatic lines at $L^* = 70$ and found DR on both *and* that a scaling fitted on one line does not transfer to the other, concluding that “diminishing returns are not constant across color space”. That is their result, not ours, and it is why we do not treat mere anisotropy as a finding: showing $M2 > M1$ would largely restate it. What their design could not do — they say so explicitly — is test whether the nonlinearity *decomposes by cone channel*, because two geodesics do not span enough directions. Our contribution begins exactly there: we supply the directions from the existing public record, formalize the hypothesis they wrote down as a nested, gauge-fixed model family, and test it against a panel of standard non-cone bases with matched complexity. The answer is negative, and the identifiability analysis says how far that negative reaches.

Bujack et al. [2025] formalize the geometry: they treat $(\mathcal{M}, \Delta E)$ as a general path-connected metric space, define the induced Riemannian metric by polarization of infinitesimal ΔE , and show that shortest *paths* coincide under the two metrics while closest *points in a set* need not. Their empirical study is a null on path coincidence, and they note that if it holds it “would also justify modeling the non-Riemannian metric ... as a Riemannian metric but with a dampening scaling function on top.” There is a live tension between that and the short paper’s conclusion that non-constancy *contradicts* a

single global dampening. Our result speaks to it directly: on the public record, the dampening that is supported *is* a single scaling of an additive cone-space metric (M1 beats M0 — decisively once locus curvature is controlled — and M2 does not beat M1), which is evidence on the CGF side of that tension — with the caveat that our chromatic data sample two isoluminant loops rather than the whole space, so we test direction-dependence, not location-dependence.

Methodologically we build on MLDS [Maloney and Yang, 2003, Knoblauch and Maloney, 2008] and its suprathreshold extension [Teti et al., 2022]; on the cone fundamentals of Stockman and Sharpe [2000], CIE [2006]; on the cone-opponent axis systems of MacLeod and Boynton [1979], Derrington et al. [1984]; and on the classical suprathreshold-scaling literature that first proposed compressive forms [MacAdam, 1963, Helm, 1964, Judd, 1979, Krantz, 1967]. The Helm and Ekman matrices themselves are among the historical exceptions to the additive-MDS consensus that Bujack et al. [2022] cite.

7 Conclusion

We asked whether perceptual diminishing returns is channel-organized and specifically cone-aligned, and tested the hypothesis in the exact form its authors posed it, on the existing public human dissimilarity record placed in a documented, normalized CIE 2006 cone space. The pipeline is anchored by a quantitative replication of the 2022 gray-axis finding (additivity ratio 0.760 [0.747, 0.772] against a published 0.757), and DR is present on the chromatic data too — indeed the model-based evidence for it becomes significant only once the model is re-expressed in coordinates that are additive along the curved stimulus loci, so it is the opposite of a curvature artifact.

The cone hypothesis, however, does not survive. The per-cone model never beats the best of five fixed, complexity-matched non-cone bases: the selection-aware held-out delta is -0.113 [-0.355, +0.007] and is negative in 16 of the 16 specifications we ran. It does not even beat a single isotropic scaling (+0.114 [-0.108, +0.327]) — and not because it is degenerate: it fits strongly distinct, strongly nonlinear per-cone scalings that simply do not generalize. The null is characterized, not merely quiet. The cone axes are very nearly degenerate on near-isoluminant loci (ΔL and ΔM project onto ridge directions 6.3° apart), yet synthetic recovery at the real design and noise level shows that per-cone scalings as distinct as the ones we simulate would nonetheless have been recovered, that the binding comparison — cones against their nearest neighbour, the opponent basis — *is* powered, and that the coarse two-channel comparison returns the same negative.

Beyond the prior work, this establishes three things. *The hypothesis Stark et al. left open is now tested*, with a gauge that makes it well posed and a control panel that makes a positive answer falsifiable. *The resolvable granularity of the public record is mapped*: the design can resolve a three-cone split of realistic size — we show this by recovery, not by assertion — and the split is not there; what the data support is an isotropic scaling of an additive cone-space metric, from which the pre- versus post-combination mechanism question cannot be adjudicated. A claim to the contrary would be over-reading. *And the between-basis differences are located*: fitting the isotropic model on every basis shows that none gains significantly from per-axis DR structure, so what separates a perceptually defined space from the cone space here is the isotropic metric it induces — a compression applied to each coordinate *before* the difference is taken — and not any channel decomposition of the difference itself.

Limitations. The chromatic evidence rests on 45 + 91 pairs from two near-isoluminant loops measured decades apart with different instructions; the CIs are correspondingly wide, and “not established” is not “absent”. The family aggregates channels additively (L_1) because that is how the hypothesis was posed; an L_2 aggregation is untested. Helm’s colorimetry is transcribed from a 1964 table and its tiles’ spectra are unavailable, so the XYZ $\rightarrow$ LMS step is a linear approximation between two observers (we report all three variants). Ekman’s adapting white is unrecorded and we test three choices. The decisive experiment this analysis calls for is not more hue-loop dissimilarity data but triads that break isoluminance while moving L and M independently — the one direction in cone space the public record does not sample.

Reproducibility. All code is in `proposal/code/`, all logs in `proposal/logs/`; the pre-registered plan is `proposal/analysis_plan.md` and the deviations from it are listed in

proposal/Attempts_log.md. Every number printed in this paper is generated from those logs by make_tables.py; none is typed by hand.

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A Full held-out panel

Table 4 lists every model we fitted, in all four primary configurations, so that nothing reported in aggregate elsewhere is hidden. Values are mean held-out log-likelihood per rating on the chromatic data, on the same folds throughout; higher is better. The coarse two-channel (K2) family is included even though only its cone member and its panel best appear in the main text. CIE LCh and CIELAB coincide in the K2 family by the CIE identity $\Delta C^{*2} + \Delta H^{*2} = \Delta a^{*2} + \Delta b^{*2}$, which is why their rows are nearly identical there; we keep both rather than silently dropping one.

Table 4: Held-out log-likelihood per rating (chromatic data) for every model in every primary configuration. Same folds, same bootstrap protocol, matched parameter counts within each family.

Model	chrom. chord	chrom. arc	joint chord	joint arc
M0 additive (no DR)	-2.230	-2.229	-2.230	-2.229
M1 isotropic DR	-2.120	-1.883	-2.109	-1.958
M2 per-cone DR	-2.006	-1.781	-2.020	-1.853
M3 cone-opponent	-2.037	-1.722	-2.065	-1.686
M3 linear sRGB	-1.982	-1.883	-1.944	-1.853
M3 CIE RGB	-1.944	-1.862	-1.938	-1.764
M3 CIELAB	-1.893	-1.893	-1.836	-1.830
M3 CIE LCh	-1.966	-1.756	-1.795	-1.871
K2 cone (achromatic + chromatic)	-2.041	-1.929	-2.028	-1.945
K2 opponent	-2.076	-1.879	-2.037	-1.908
K2 sRGB	-2.002	-1.915	-2.003	-1.880
K2 CIE RGB	-2.055	-1.936	-1.958	-1.887
K2 CIELAB	-1.857	-1.863	-1.842	-1.871
K2 CIE LCh	-1.857	-1.863	-1.842	-1.869

Table 5 gives the coarse-family analogue of the per-control pairwise deltas that Table 1 reports for the M3 panel: the cone-defined coarse split is level with the opponent, sRGB and CIE RGB splits and behind the two perceptually defined ones, with every interval wide.

Table 5: Coarse (two-channel) family: paired held-out delta of the cone split against each matched non-cone control, with 95% cluster-bootstrap CIs. CIELAB and CIE LCh coincide here by the CIE identity noted above.

Coarse control	K2 cone — control (chromatic, chord)
K2 opponent	+0.035 [-0.089, +0.188]
K2 sRGB	-0.039 [-0.217, +0.159]
K2 CIE RGB	+0.014 [-0.196, +0.249]
K2 CIELAB	-0.184 [-0.413, +0.098]
K2 CIE LCh	-0.184 [-0.413, +0.098]